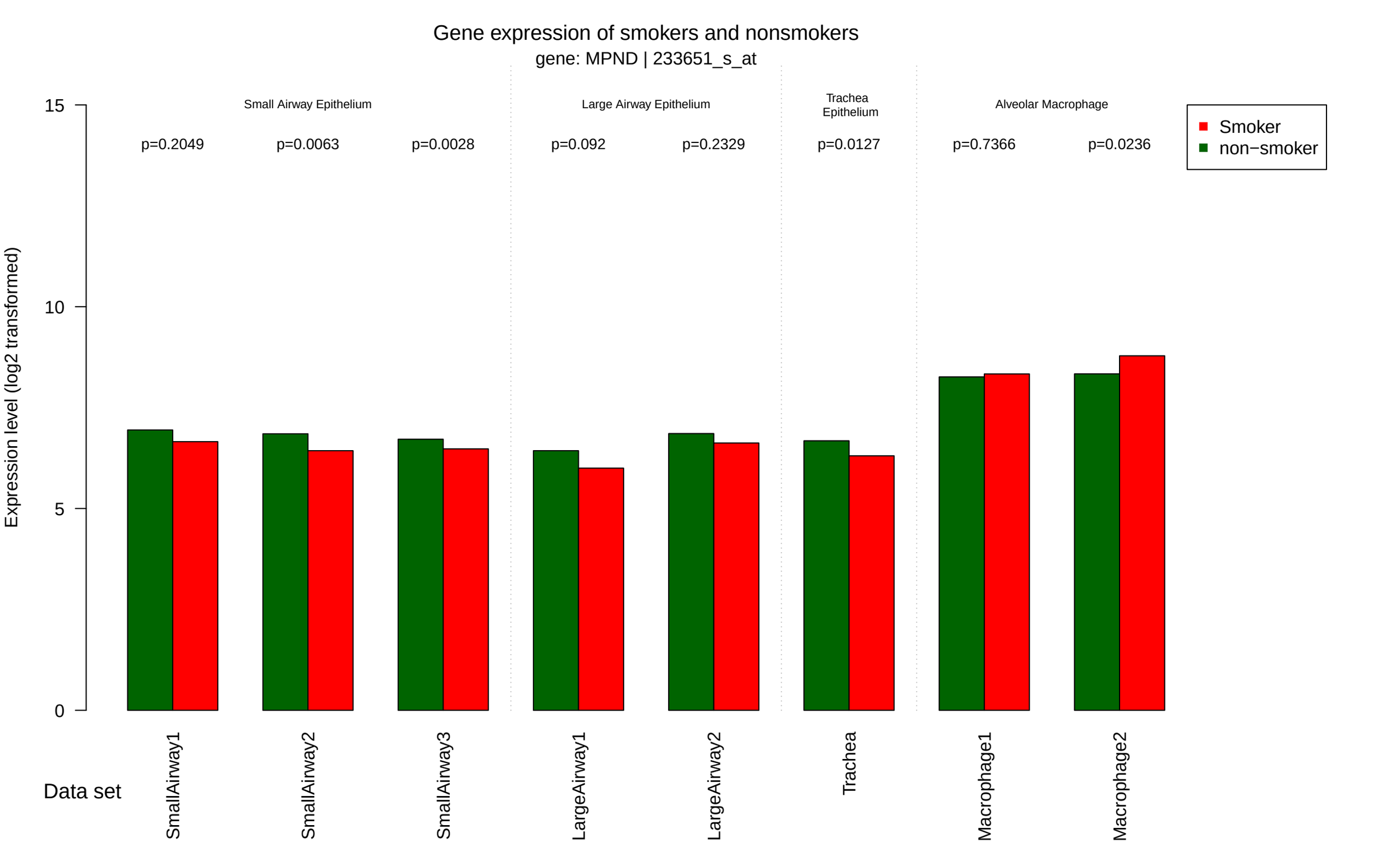
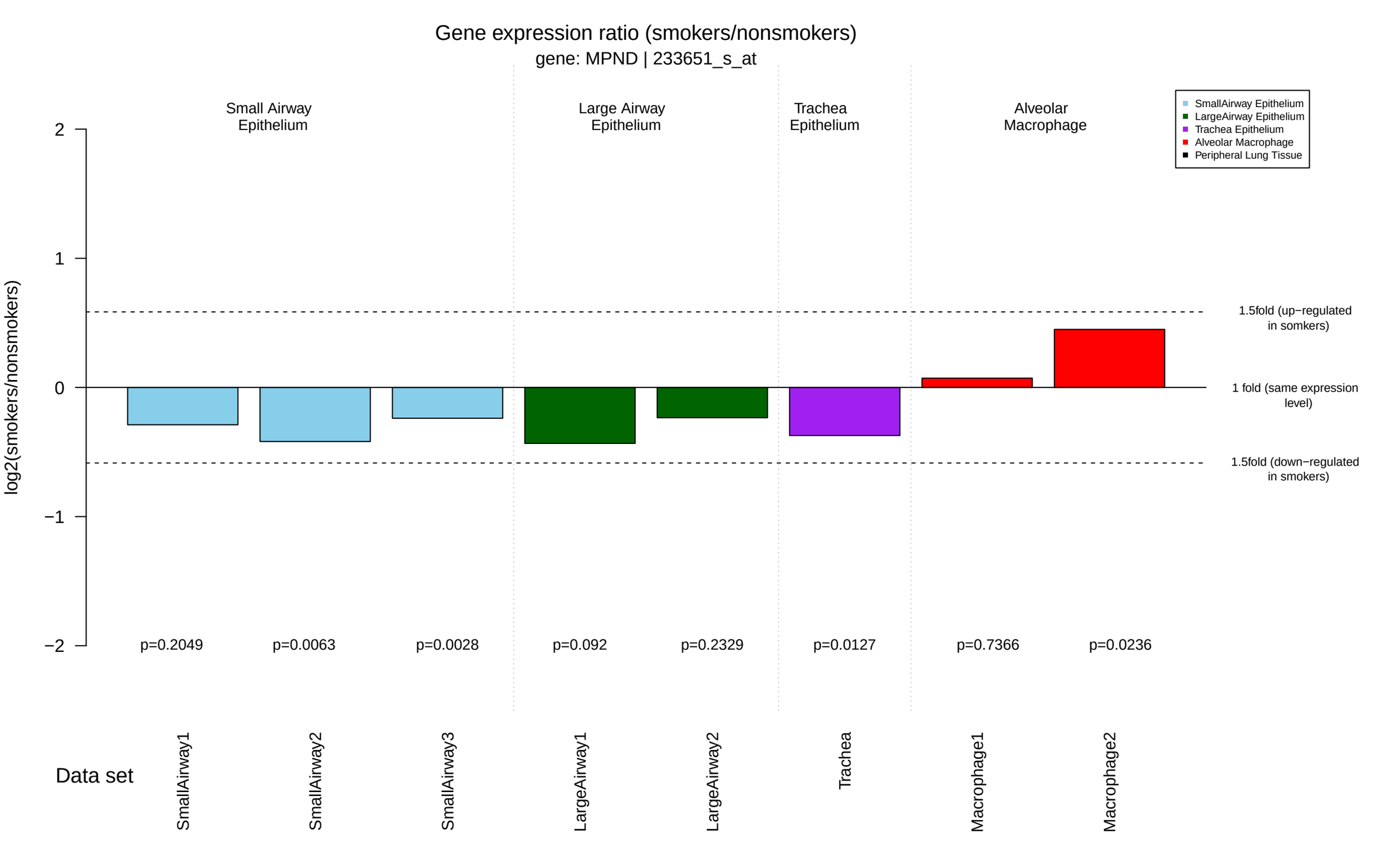
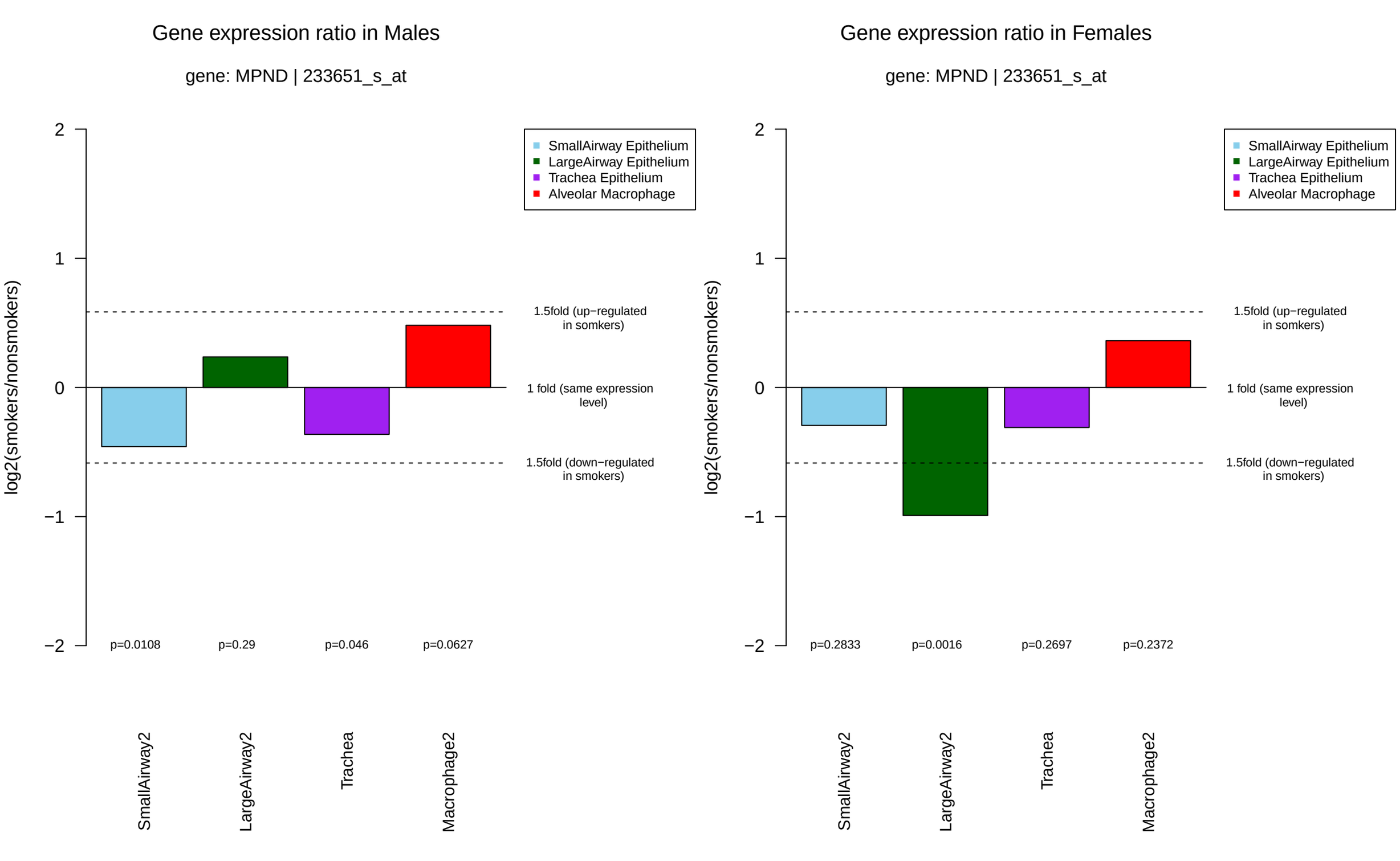
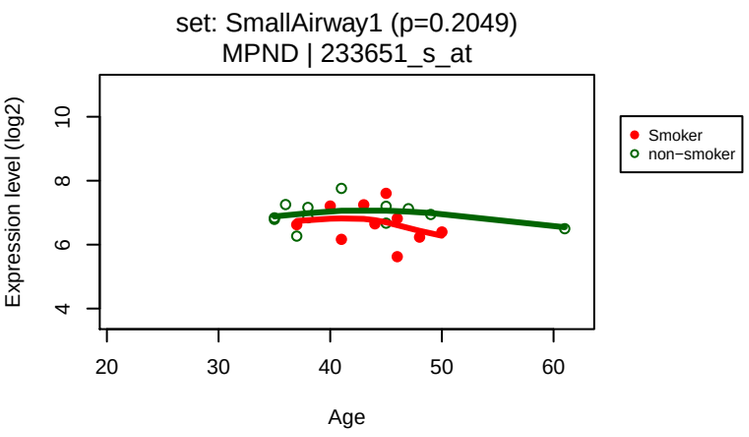
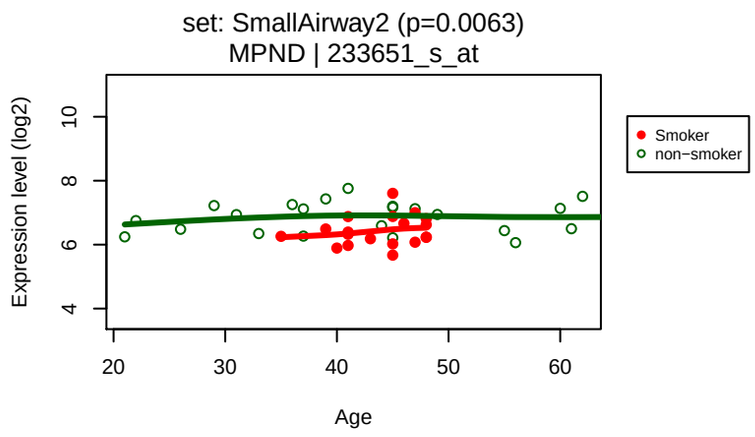
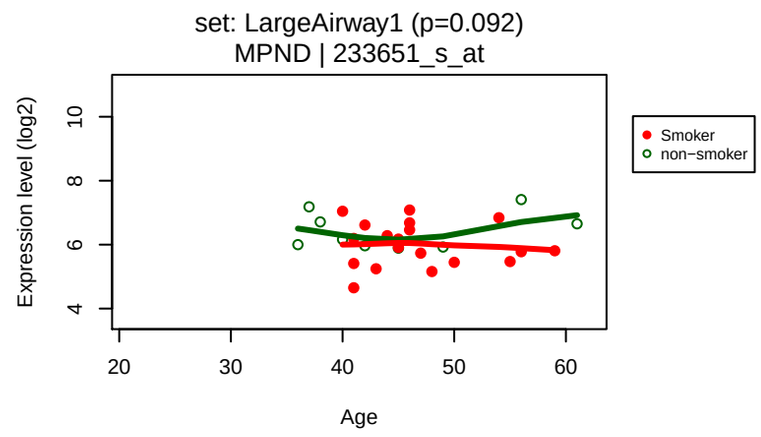

Gene expression of smokers and nonsmokers
gene: MPND | 233651_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2049
p=0.0063
p=0.0028
p=0.092
p=0.2329
p=0.0127
p=0.7366
p=0.0236
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

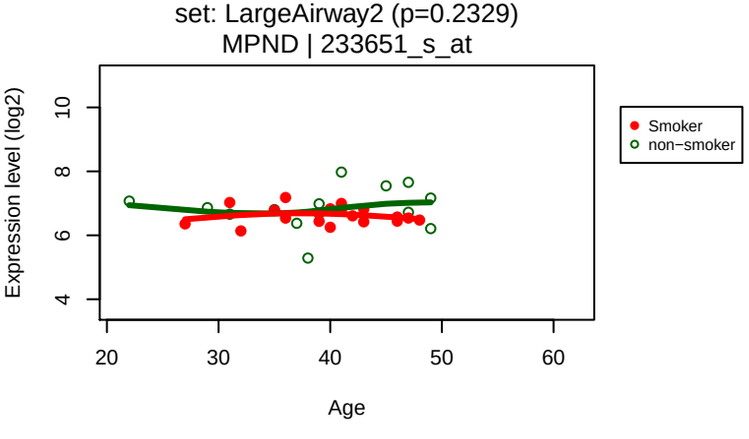

Gene expression ratio (smokers/nonsmokers)
gene: MPND | 233651_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2049
p=0.0063
p=0.0028
p=0.092
p=0.2329
p=0.0127
p=0.7366
p=0.0236
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

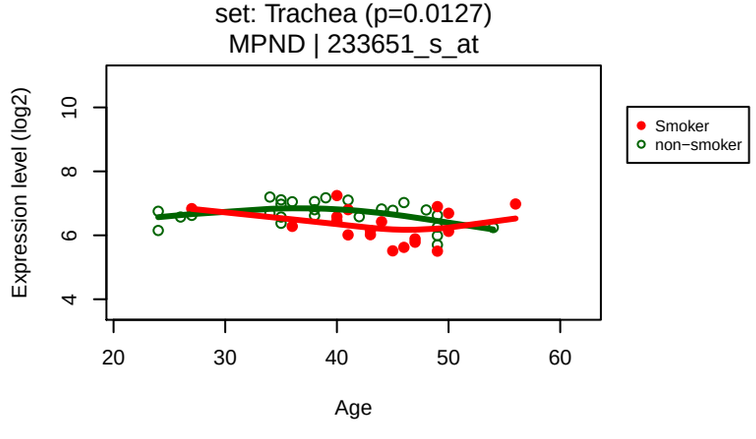

Gene expression ratio in Males
gene: MPND | 233651_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0108
p=0.29
p=0.046
p=0.0627
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MPND | 233651_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2833
p=0.0016
p=0.2697
p=0.2372
SmallAirway2
LargeAirway2
Trachea
Macrophage2

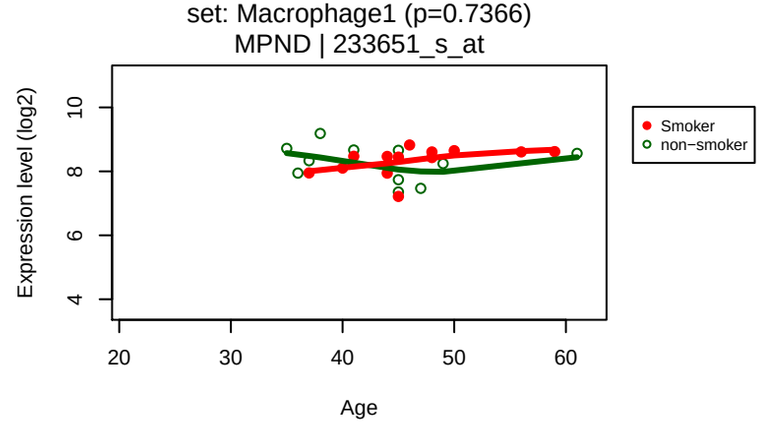

set: SmallAirway1 (p=0.2049)
MPND | 233651_s_at
Expression level (log2)
Age
Smoker
non−smoker

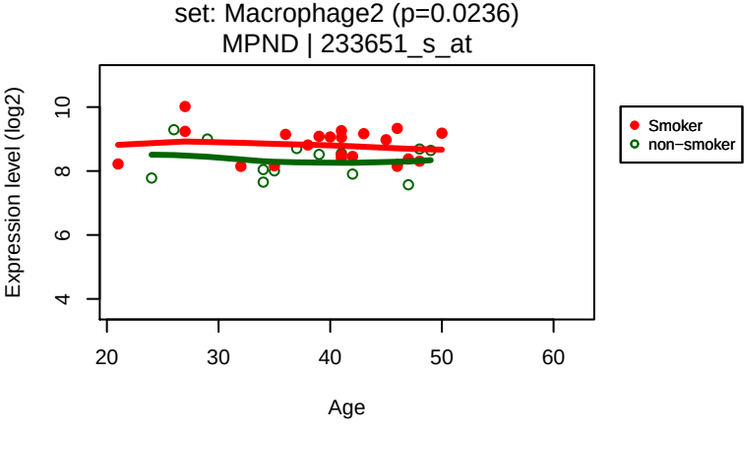

set: SmallAirway2 (p=0.0063)
MPND | 233651_s_at
Expression level (log2)
Age
Smoker
non−smoker

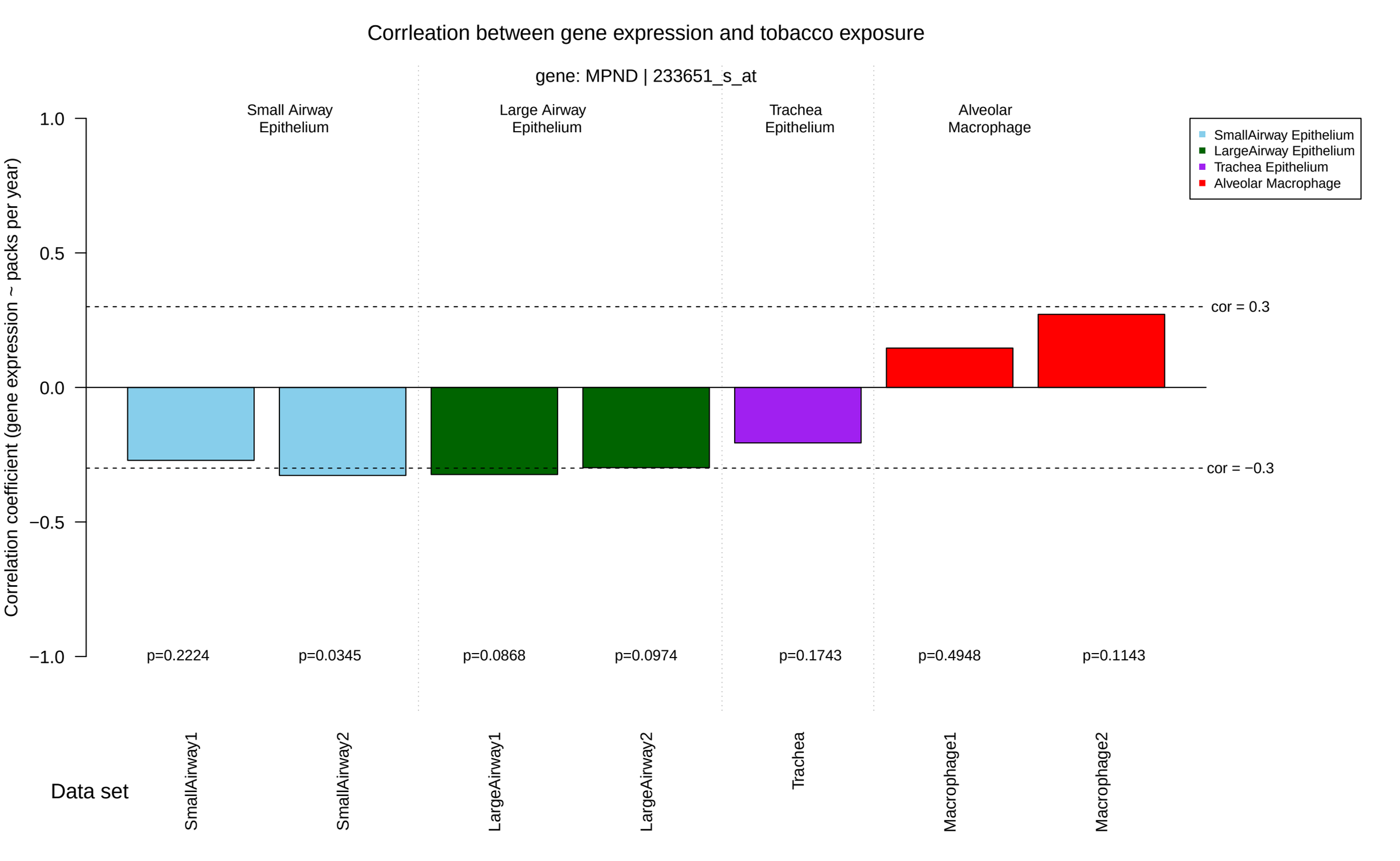

set: LargeAirway1 (p=0.092)
MPND | 233651_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2329)
MPND | 233651_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0127)
MPND | 233651_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7366)
MPND | 233651_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0236)
MPND | 233651_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MPND | 233651_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2224
p=0.0345
p=0.0868
p=0.0974
p=0.1743
p=0.4948
p=0.1143
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2